Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 34: Order Matters

Permutations

Chartrand & Zhang, Section 8.4, pp. 292–302

Name: _____ Date: _____

Where we are

Last week's principles solve any counting problem in principle — but writing $10 \cdot 9 \cdot 8 \cdot 7 \cdot 6$ every time is tedious. Today we package the two most common patterns into formulas. Nothing new is being assumed; both come straight from the Multiplication Principle.

By the end of today you should be able to

1. Define an r-permutation and compute $P(n, r)$.
2. Derive $P(n, r) = \dfrac{n!}{(n-r)!}$ from the Multiplication Principle.
3. Recognise when a problem is asking for an ordered selection.

1. Factorials

Factorial

For a positive integer n, $n! = n \cdot (n-1) \cdots 2 \cdot 1$, and by convention $0! = 1$.

n	0	1	2	3	4	5
$n!$	1	1	2	6	24	120

Why $0! = 1$? Not a whim. There is exactly one way to arrange nothing — the empty arrangement — and the convention keeps every formula below working at the edges.

2. Permutations

Definition

A **permutation** of a set is an arrangement of all its elements in order. An **r-permutation** is an ordered list of r of its elements. The number of r-permutations of an n-element set is written

$P(n, r)$.

Derive the formula. Building an ordered list of r items from n:

Position	1st	2nd	3rd	$\cdots$	rth
Choices	n	$n-1$	$n-2$	$\cdots$	$n-r+1$

By the Multiplication Principle,

$$P(n, r) = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}.$$

The second form comes from multiplying and dividing by $(n-r)!$ — the factors we did *not* use.

Count	**Answer**
Ways to seat 5 people in a row	$P(5,5) = 5! = 120$
Gold, silver, bronze among 10 runners	$P(10,3) = 10 \cdot 9 \cdot 8 = 720$
Four-letter strings from COMPUTER, no repeats	$P(8,4) = 8 \cdot 7 \cdot 6 \cdot 5 = 1680$

Two edge cases the convention handles

$P(n,n) = \dfrac{n!}{0!} = n!$ — arranging everything. And $P(n,0) = \dfrac{n!}{n!} = 1$ — the empty list, counted once. Both work only because $0! = 1$.

3. Spotting a permutation problem

The question to ask

Does rearranging the chosen items give a different outcome?
If yes — gold/silver/bronze, passwords, seatings, rankings — it is a permutation. If no — a committee, a hand of cards, a subset — it is something else, and that is Wednesday.
Signal words for *ordered*: arrange, rank, schedule, in a row, first/second/third, password.

Classify each.

Problem	Ordered?
Choosing a president, treasurer, and secretary from 12	Yes — roles differ
Choosing a 3-person committee from 12	No — no roles
Dealing 5 cards to one player	No — a hand is a set
Arranging 5 books on a shelf	Yes

Exit ticket

1. Compute $P(7, 3)$, showing the product.

2. Is "choose 4 students to receive identical prizes" a permutation problem? Explain.

Before Wednesday

Read Section 8.4, pp. 296–302 — combinations.

Self-check (not collected). Section 8.4, Exercises 1, 3, and 5 — odd-numbered, answers in the back.

Coming up

Homework 12 is due Friday, November 13, 11:59 p.m.
Quiz 11 is Monday, November 16 — Section 8.4.

Questions I still have